South African Journal of Botany 124 (2019) 223–227

Contents lists available at ScienceDirect

South African Journal of Botany

journal homepage: www.elsevier.com/locate/sajb

Green synthesis of titanium dioxide nanoparticles using *Azadirachta indica* leaf extract and evaluation of their antibacterial activity

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ARTICLE INFO

Article history:
Received 14 April 2019
Received in revised form 9 May 2019
Accepted 14 May 2019
Available online 28 May 2019

Edited by A Andreucci

Keywords:
Green synthesis
Nanoparticles
Azadirachta indica
Antibacterial activity

ABSTRACT

The present study deals with the synthesis of titanium dioxide nanoparticles using *Azadirachta indica* leaf extract. FTIR study reveals the presence of terpenoids, flavonoids and proteins considered responsible for the formation and stabilization of titanium dioxide nanoparticles while the XRD studies revealed the crystalline nature of titanium dioxide nanoparticles. Transmission electron microscopy images showed the spherical shape and size ranged from 15 to 50 nm. SEM study revealed that TiO_2 nanoparticles were spherical in shape and their size ranged from 25 to 87 nm. The antibacterial activity of synthesized TiO_2 nanoparticles and TiO_2 compound was assayed against *Escherichia coli, Bacillus subtilis, Salmonella typhi* and *Klebsiella pneumoniae.* The results of the present study showed that TiO_2 nanoparticles inhibited the growth of all the tested microorganisms. The antibacterial effect is more pronounced in case of TiO_2 nanoparticles as compared with TiO_2 compound. The lowest MIC (minimum inhibitory concentration) value i.e. 10.42 μg/mL of nanoparticles was observed against *Salmonella typhi* and *Escherichia coli* whereas lowest MBC value i.e. 83.3 μg/mL was observed against *Klebsiella pneumoniae.*

1. Introduction

Prevention of infectious diseases through control of pathogens has been a major challenge across the globe. Numerous antibiotics and antifungal formulations are available for control of diseases caused by microorganisms. Because of increased use of antibiotics, antibiotic resistance is becoming a common problem. Bacteria have become resistant to antibiotics. Thus there is a need for exploring alternative strategies for spread of infectious diseases. In the present scenario, nanoscale materials have emerged up as novel antimicrobial agents owing to their high surface area to volume ratio and its unique chemical and physical properties (Kim et al., 2007).

Nanoparticles (10^{-9} m) are defined as small objects that behave as a whole unit in terms of its transport and properties. There are reports on antimicrobial activity of nanoparticles such as Ag, Au, MgO, CuO, Al, TiO_2 etc. which are effective against different drug resistant bacterial, viral and fungal strains (Rai and Bai, 2011). Therefore, a nanoparticle has potential to serve as an alternative to antibiotics and to control microbial infections caused by *Staphylococcus aureus*, *Staphylococcus epidermidis*, *Klebsiella pneumoniae* and *Salmonella typhi* (Luo et al., 2007). Amongst the metal oxide nanoparticles, TiO_2 nanoparticles are more remarkable nanomaterial due to their photocatalytic properties, chemical stability and non-toxicity. It has enormous uses in cosmetic industry, solar cells, electorochemical devices, pollution control and antibacterial coatings (Amarnath et al., 2013). *Azadirachta indica* (neem) have exceptional therapeutic properties. There are a number of biologically active phytoconstituents in *Azadirachta indica* leaves which include flavonoids, alkaloids, terpenoids and polyphenols which can be used in the reduction of titanium dioxide ions (Dubey et al., 2009).

Numerous approaches have been used to synthesize nanoparticles and toxic chemicals that are used in physical and chemical methods may reside in the nanoparticles formed which may prove hazardous to the environment and organisms in the field of their application (Dhandapani et al., 2014). Green synthesis is an efficient technique which includes synthesis through plants, bacteria, fungi, algae, actinomycetes etc. Green approach is an environment friendly, cost-effective, biocompatible and safe (Zahir et al., 2014). Nanoparticles synthesized from this approach produce more catalytic activity and limit the use of expensive and toxic chemicals (Kumar et al., 2017). In recent years, the synthesis of nanoparticles using green approach has been increased. There are reports on the synthesis of TiO_2 nanoparticles using different plant extracts (Nasrollahzadeh and Sajadi, 2015, 2016; Rostami-Vartooni et al., 2016; Nasrollahzadeh et al., 2016; Maham et al., 2017). The present study is endeavored to use green approach

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https://doi.org/10.1016/j.sajb.2019.05.024
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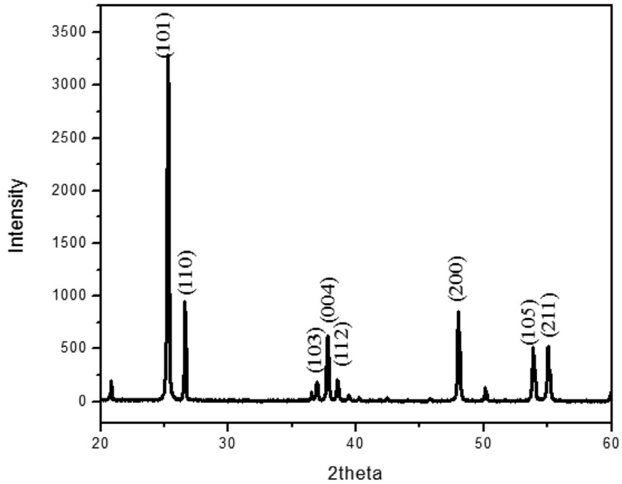

Fig. 1. X-ray Diffraction spectra of TiO_2 nanoparticles.

to synthesize titanium dioxide nanoparticles using *Azadirachta indica* leaf extract and explore the antibacterial potential of synthesized nanoparticles.

2. Materials and methods

2.1. Materials

Healthy leaves of *Azadirachta indica* were collected from Chandigarh, India. For antimicrobial activity, bacterial pathogens viz. *E. coli, Klebsiella pneumoniae, Bacillus subtilis, Staphylococcus aureus* and *Salmonella typhi* were obtained from Post Graduate Institute of Medical Education and Research (PGIMER), Chandigarh, India.

2.2. Preparation of Azadirachta indica leaf broth

Briefly 20 g of finely cut fresh *Azadirachta indica* leaves were thoroughly washed with distilled water and then boiled in 100 mL of sterile double distilled water for 30 min in water bath. The resulting extract was then filtered to remove particulate matter and was stored at 4 °C for further use.

2.3. Green synthesis of TiO_2 nanoparticles

Green synthesis of TiO_2 nanoparticles was performed as described earlier by Krishnasamy et al. (2015) with modifications. Briefly, 10 mL of *Azadirachta indica* leaf broth was added to 90 mL of 5 mM of aqueous titanium dioxide for synthesis of nanoparticles. The titanium dioxide nanoparticles were purified by repeated centrifugation at 7826 $\times \underline{g}$ for 20 min using double distilled water and pellet was dispersed in the deionized water. Finally, heat treatment was given at 60 °C for 1 h and stored for further use.

2.4. Characterization of nanoparticles

Synthesized titanium dioxide nanoparticles were characterized using SEM, TEM, FTIR spectroscopy and XRD. Morphology and size and of the synthesized TiO_2 nanoparticles were analyzed by field emission scanning electron microscope (NOVA NANOSEM 450, FEI, USA). Transmission electron microscopy (FP 5022/22-Tecnai G2 20 S-TWIN, FEI, USA) was performed for characterizing the size and shape of TiO_2 nanoparticles. The FTIR spectroscopy study of synthesized nanoparticles was performed using Cary, 630 FTIR (Agilent Technologies, India) in the spectrum range from 4000 to 400 cm^{-1}. X-Ray diffraction analysis of TiO_2 nanoparticles was carried out using SmartLab 9 kW rotating anode x-ray diffractometer (Rigaku Corporation) in the diffraction angles (2θ) from 20^0 to 80^0 (scanning range) at 2^0/min scanning rate.

2.5. TiO_2 nanoparticles preparation

A stock solution was prepared by suspending the nanoparticles in methanol to yield a final conc. of 10 mg/mL. The stock solution was sonicated for 15 min and every assay was done within 1–2 h of sonication. The suspension was kept at 4 °C.

2.6. Antimicrobial activity of TiO_2 NPs

Antimicrobial activity of TiO_2 nanoparticles was performed using modified well diffusion assay (NCCLS, 1993).

2.6.1. Preparation of inoculums

Briefly 4–5 well isolated colonies of the same morphological type were selected from an agar plate culture and transferred into a tube containing 5–6 mL of Nutrient broth medium. The broth culture was

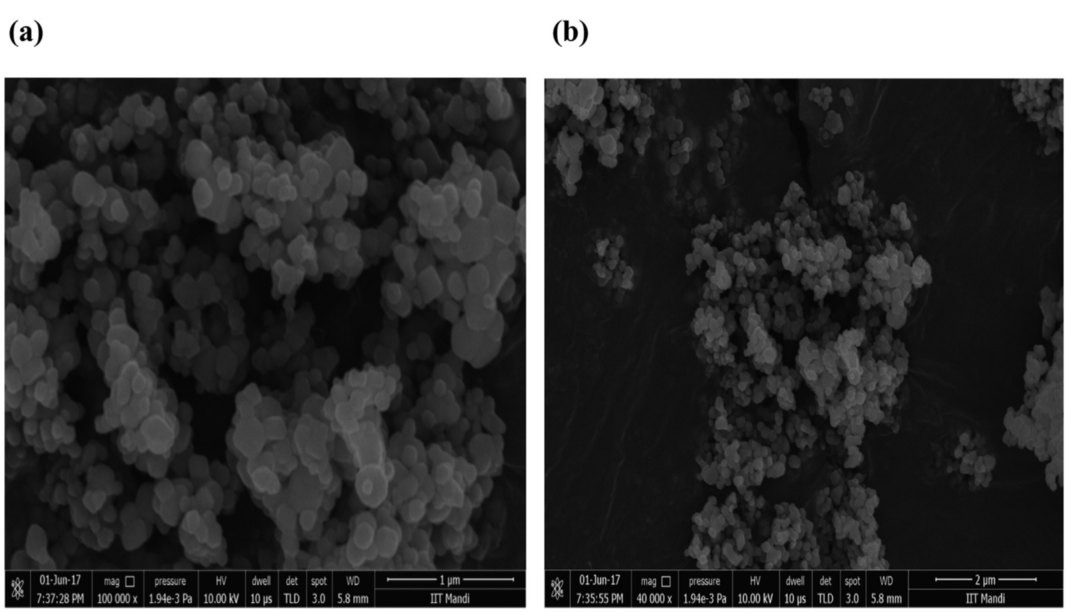

Fig. 2. SEM images of TiO_2 nanoparticles at (a) 1 µm and (b) 2 µm.

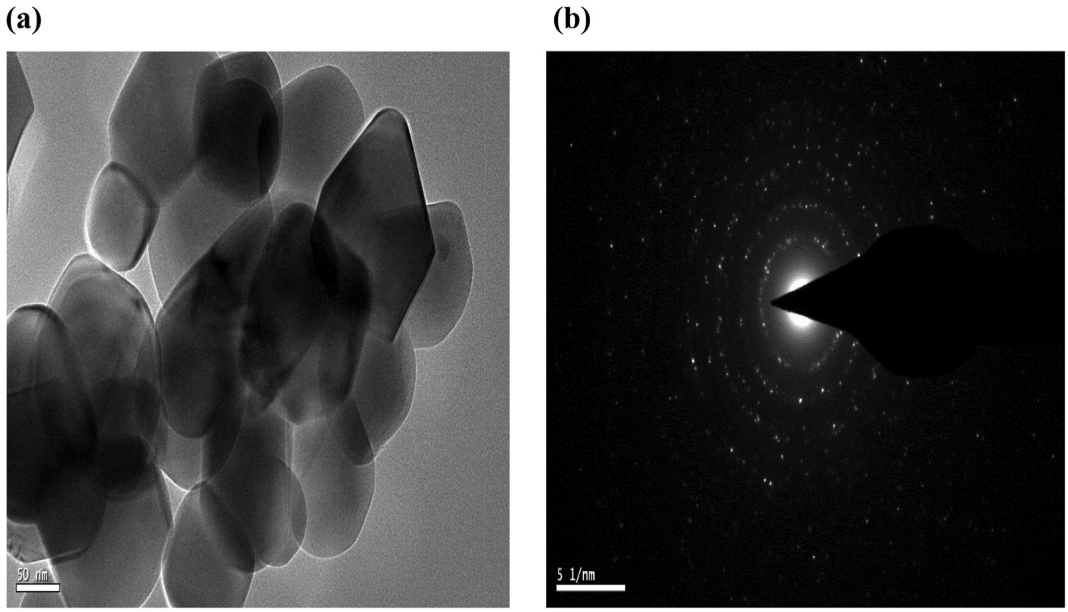

Fig. 3. (a) TEM image of synthesized TiO_2 nanoparticles at 50 nm scale. (b) SAED image of TiO_2 nanoparticles.

incubated at 37 °C for 24 h. The turbidity of inoculum was compared with 0.5 McFarland standards containing 1–2 × 10^8 CFU/mL.

2.6.2. Well diffusion assay

Mueller Hinton agar (MHA) plates were prepared. A sterile cotton swab was dipped into each inoculum and swab streaked over the entire surface of respective MHA plates and then with help of cork borer wells were made in the plates. Different conc. (66, 133 and 200 μg/mL *w*/*v*) of TiO_2 nanoparticles was loaded into the wells and plates were incubated at 37 °C for 24 h. Methanol was used as negative control and tetracycline (25 μg/mL *w*/*v*) was used as positive control. After incubation, the zone of inhibition was measured using Hi antibiotic zone scale.

2.6.3. Minimum inhibitory concentration (MIC) of TiO_2 nanoparticles

MIC of titanium dioxide nanoparticles was performed using broth micro dilution method (CLSI, M7-A7, 2006). Bacterial strains were grown overnight in nutrient broth at 37 °C. In this test, double strength Muller–Hinton broth (MHB) was used and 2X solution of TiO_2 nanoparticles were prepared in methanol. Each well of 96-well microtiter plate was aliquoted with 100 μL of MHB. Briefly, 100 μL of Titanium dioxide nanoparticles (2X) were added into the first well and serial dilutions were made in the conc. range of 200–0.78 μg/mL up to the 9th well. After that, 10 μL bacterial suspension (1–2 × 10^8 CFU/mL) was added to each well. The 10th well was taken as growth control well (containing MHB and inoculum), 11th well as negative control (Containing MHB, inoculum and methanol) and 12th well was taken as positive con-

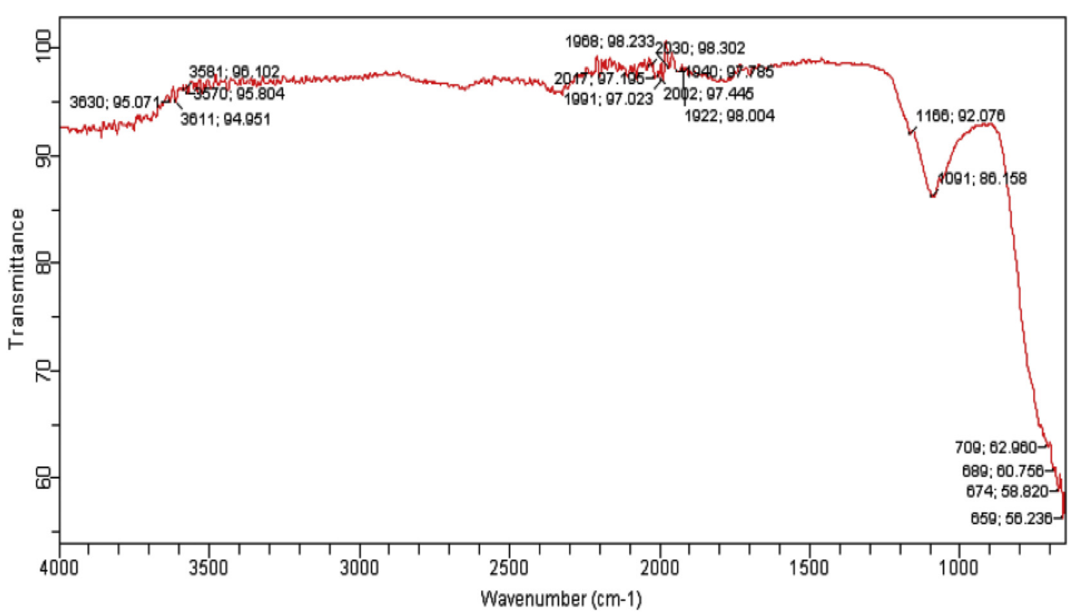

Fig. 4. FT-IR spectrum of TiO_2 nanoparticles.

Table 1
Inhibition zone diameter of TiO_2 Compound and TiO_2 nanoparticles against different bacterial pathogens.

Bacterial strains	*IZD (mm) for TiO_2 Compound ± SD Conc. (μg/ml)			*IZD (mm) for TiO_2 nanoparticles ± SD Conc. (μg/ml)		
	66	133	200	66	133	200
S. aureus	14.3 ± 1.46	17 ± 1.16	19.33 ± 1.45	18 ± 1.15	21.67 ± 1.45	27 ± 1.15
K. pnuemoniae	10 ± 1.15	13 ± 1.15	20.67 ± 1.45	13 ± 1.15	17 ± 1.15	25.33 ± 1.20
B. subtilis	10 ± 1.15	13.33 ± 1.2	15 ± 1.56	14.67 ± 2.03	18.33 ± 1.45	24 ± 1.73
S. typhi	11 ± 1.20	11.33 ± 1.76	14 ± 1.45	14 ± 1.53	18.33 ± 1.76	21.63 ± 1.45
E.coli	11 ± 1.52	14.67 ± 1.45	17.33 ± 1.45	15 ± 1.15	21.33 ± 1.45	27.67 ± 0.76

IZD = Inhibition zone diameter, SD = Standard deviation

trol (containing MHB, inoculum and tetracycline). The lowest concentration showing visual inhibition of growth was considered as the MIC of the TiO_2 nanoparticles for the respective organism.

2.6.4. Minimum bactericidal concentration (MBC) of TiO_2 nanoparticles

Briefly, 10 μL of suspension from each well of microtiter plate were inoculated on nutrient agar plate and incubated at 37 °C for 24 h. The same was repeated for control wells (positive and negative). The nanoparticles concentration causing bactericidal effect was selected based on absence of colonies on the agar plates.

2.7. Statistical analysis

All the experiments were done in triplicates, average and standard error were determined using GraphPad Prism 5.0.

3. Results and discussion

3.1. Synthesis and characterization of TiO_2 nanoparticles

White colored titanium dioxide nanoparticles were prepared using the extract of *Azadirachta indica* and characterized using XRD, SEM, TEM and FT-IR spectroscopy. XRD patterns of green synthesized TiO_2 nanoparticles were observed at 25.36, 26.54, 37.05, 37.78, 38.54, 48.12, 54.02 and 55.04° can be attributed to the 101, 110, 103, 004, 112, 200, 105 and 211 crystalline anatase and rutile structures of synthesized titanium dioxide nanoparticles (Fig. 1). These results were confirmed using Joint Committee on Power Diffraction Standards (JCPDS) No. 21-1272 and 21-1276. The average crystalline size of synthesized nanoparticles was 41.9 nm calculated using Scherrer's equation ($D = K\lambda/\beta cos\theta$). λ is the wavelength of x-ray radiation ($CuK\alpha = 0.15406$ nm), K is a constant taken as 0.89, β is the line width at half maximum height (FWHM) of peak and θ is the diffracting angle. The average crystalline size of synthesized nanoparticles is 15–45 nm calculated by Scherrer's formula based on the XRD pattern (Patterson, 1939).

The surface morphology and size of the TiO_2 nanoparticles were examined using scanning electron microscopy (SEM) which revealed that TiO_2 nanoparticles were spherical and their size was in range from 25 to 87 nm (Fig. 2). TEM studies showed that the nanoparticles are spherical in shape with smooth surfaces and the size in range 15–45 nm which is in close covenant with that estimated by Scherrer's formula based on the XRD pattern (Fig. 3 a). The SAED (Selected area electron diffraction) image reveals the crystalline structure complexity and bright small spots making up the ring show the polynano crystalline nature of the sample (Fig. 3 b). The size of the biosynthesized TiO_2 nanoparticles in SEM analysis was higher than that of TEM analysis due to inconsistent calibration of magnification, secondary electron emission increase and sample preparation (Tuoriniemi et al., 2014).

The FTIR spectrum of biosynthesized titanium dioxide nanoparticles showed peaks at 3581.96, 1166.92, 1091.86, and 709.62 (Fig. 4). The peak at 3581.96 corresponds to the O–H stretching of alcohols and phenolic compounds, peak at 1166.92 attributed to C=C groups of aromatic rings. The peak at 1091.86 denotes the C=O vibrations of carboxylic acid and peak at 709.62 correspond to the Ti–O–Ti stretching vibration of titanium dioxide nanoparticles. Similar observations were also reported by Vasconcelos et al. (2011) and Yilmaz et al. (2011).

Table 2
MIC and MBC of titanium dioxide nanoparticles against bacterial pathogens.

Bacterial strains	TiO_2 nanoparticles	
	MIC(μg/ml)	MBC (μg/ml)
S. aureus	20.83	133.3
K. pnuemoniae	16.66	83.33
B. subtilis	25	166.67
S. typhi	10.42	100
E. coli	10.42	166.67

MIC = Minimum Inhibitory Concentration, MBC = Minimum Bactericidal Concentration

3.2. Antibacterial activity of TiO_2 nanoparticles

Titanium dioxide nanoparticles synthesized using *Azadirachta indica* leaf extract were found effective against microbial pathogens viz. *Escherichia coli, Staphylococcus aureus, Bacillus subtilis, Salmonella typhi* and *Klebsiella pneumoniae.* All the pathogens were found susceptible against both titanium dioxide compound and synthesized titanium dioxide nanoparticles. Synthesized titanium dioxide nanoparticles were more effective as compared with titanium dioxide compound. Titanium dioxide nanoparticles showed maximum activity against *E. coli* with inhibition zone diameter 27.67 ± 0.76 mm at 200 μg/mL. Titanium dioxide compound showed highest zone of inhibition i.e. 20.67 ± 1.45 mm at 200 μg/mL against *Klebsiella pneumoniae* (Table 1). The results of antibacterial assay were analyzed in the light of existing literature. The antibacterial activity of TiO_2 nanoparticles synthesized using *Psidium guajava* extract was evaluated by Santhoshkumar et al. (2014). They reported that nanoparticles were found the most effective against *Staphylococcus aureus* and *Escherichia coli.* The results of MIC and MBC of synthesized titanium dioxide nanoparticles correlate well with those of well diffusion assay which is a qualitative test. The lowest MIC value i.e. 10.42 μg/mL of nanoparticles was reported against *Salmonella typhi* and *Escherichia coli* whereas lowest MBC values i.e. 83.3 μg/mL were reported against *Klebsiella pneumoniae* (Table 2).

4. Conclusion

Green synthesis of titanium dioxide nanoparticles was achieved because of the presence of terpenoids, flavonoids and proteins in *Azadirachta indica* as these bioactive compounds were responsible for the synthesis of these nanoparticles. Synthesis using green approach is simple, inexpensive and eco-friendly process which reduces the use of toxic chemicals. Synthesized titanium dioxide nanoparticles exhibited broad spectrum antimicrobial activity against vast range of pathogens. In the current scenario, keeping in view the problem of multi drug resistance in bacteria, one of the most promising and novel antimicrobial agents could be the nanoparticles.

Acknowledgements

The authors are thankful to Mr. Madan, Faculty of Basic Sciences, Shoolini University, Solan, India for his help in characterization of nanoparticles and Indian Institute of Technology, Mandi, India for providing necessary facilities to carry out the characterization work in the present study.

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